



Object-Oriented Software Development

Lecture 6 - Extra

Drawing Sequence Diagram

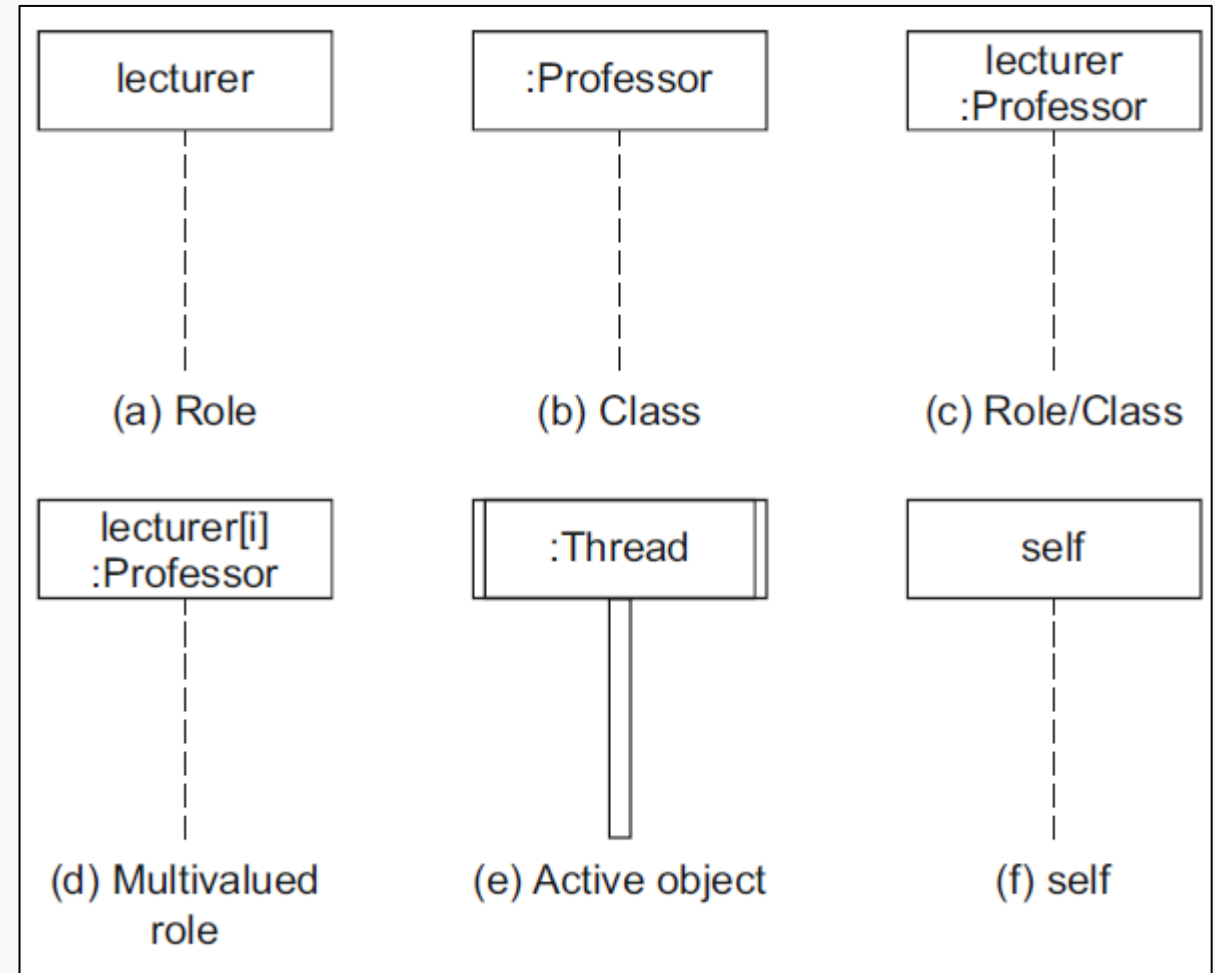
(d) Multivalued role:

- can be connected to more than one object (multivalued role).
- Represents a selector.
- is specified in square brackets between the role name and the colon.
- the selector is simply a variable that acts as an index

(e): Active object:

- are used to model processes and threads.

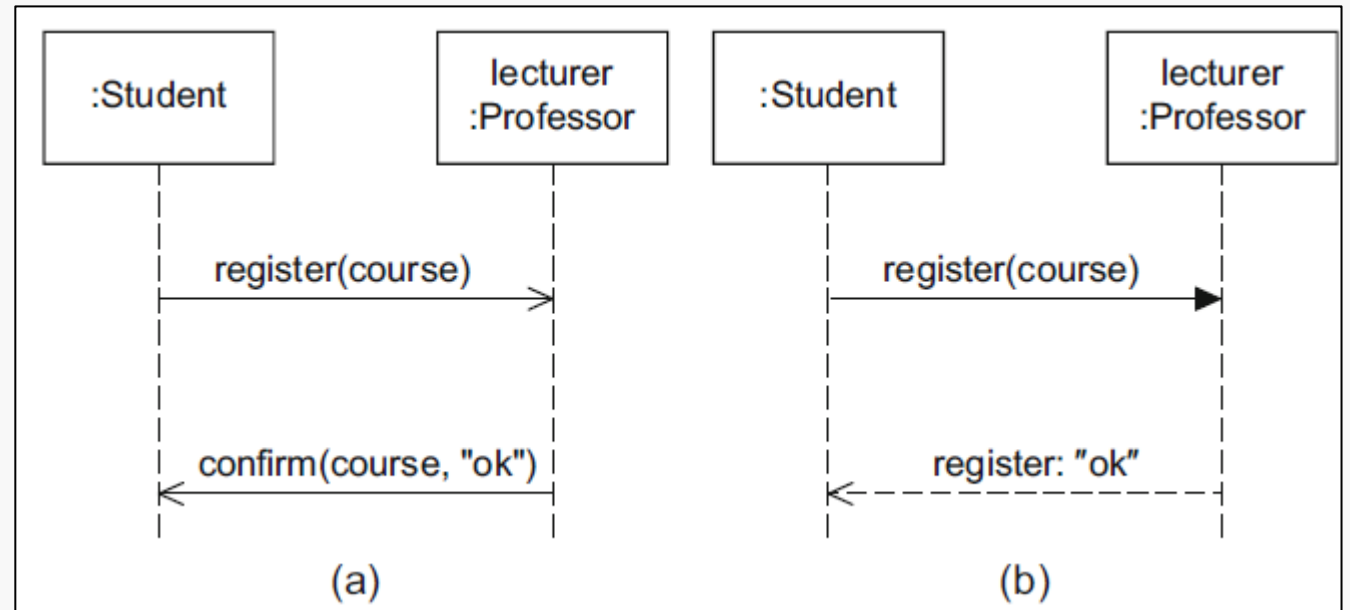
(f) Self: is needed when a class spans a certain interaction context and is itself involved in the interaction.



Types of lifelines

Drawing Sequence Diagram

- Asynchronous messages can have a reply if needed in business, and the reply is specified as a solid line with an open head.
- If the content of the response message and the point at which the response message is sent and received are clear from the context, then the response message may be omitted in the diagram

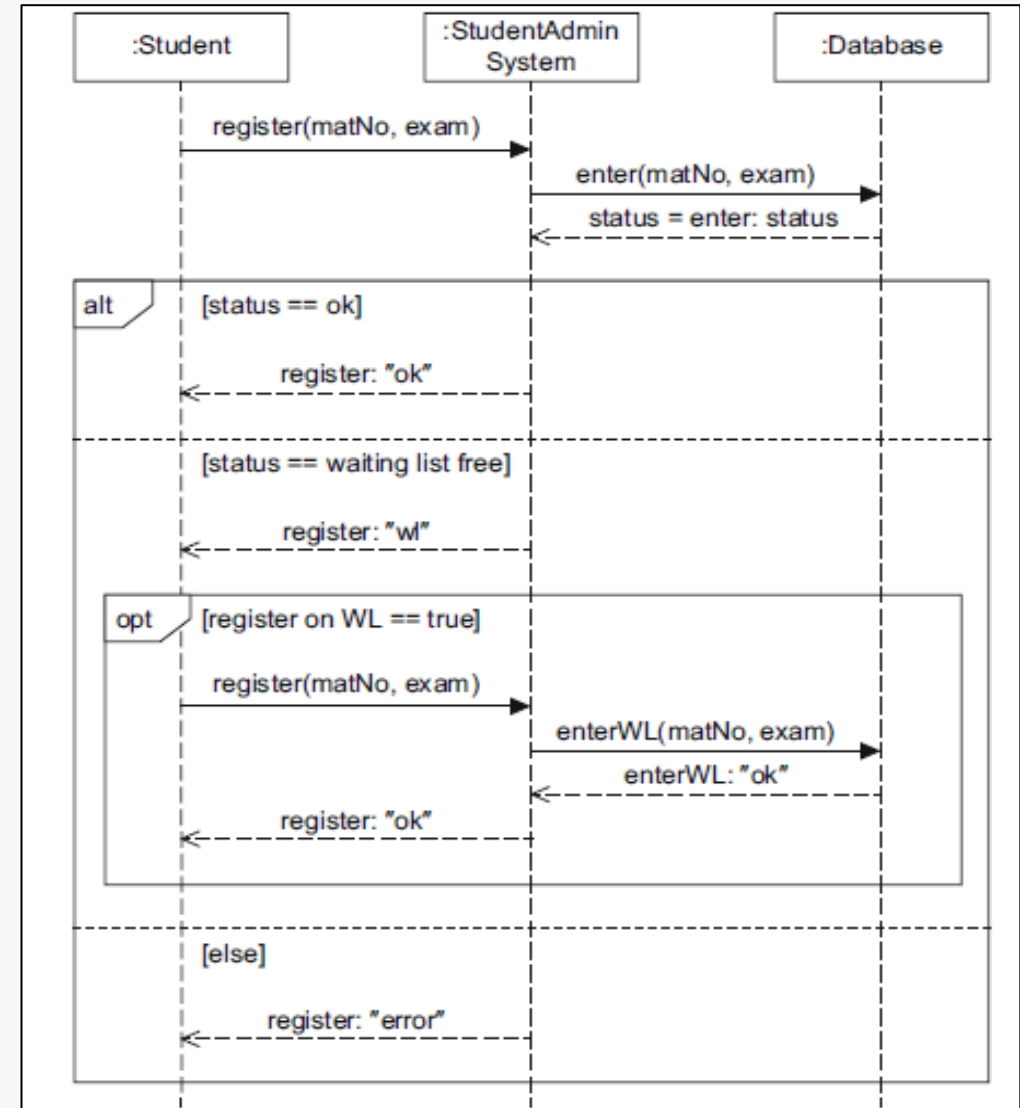


Examples of (a) asynchronous and (b) synchronous Communication.

Drawing Sequence Diagram

Branches and Loops:

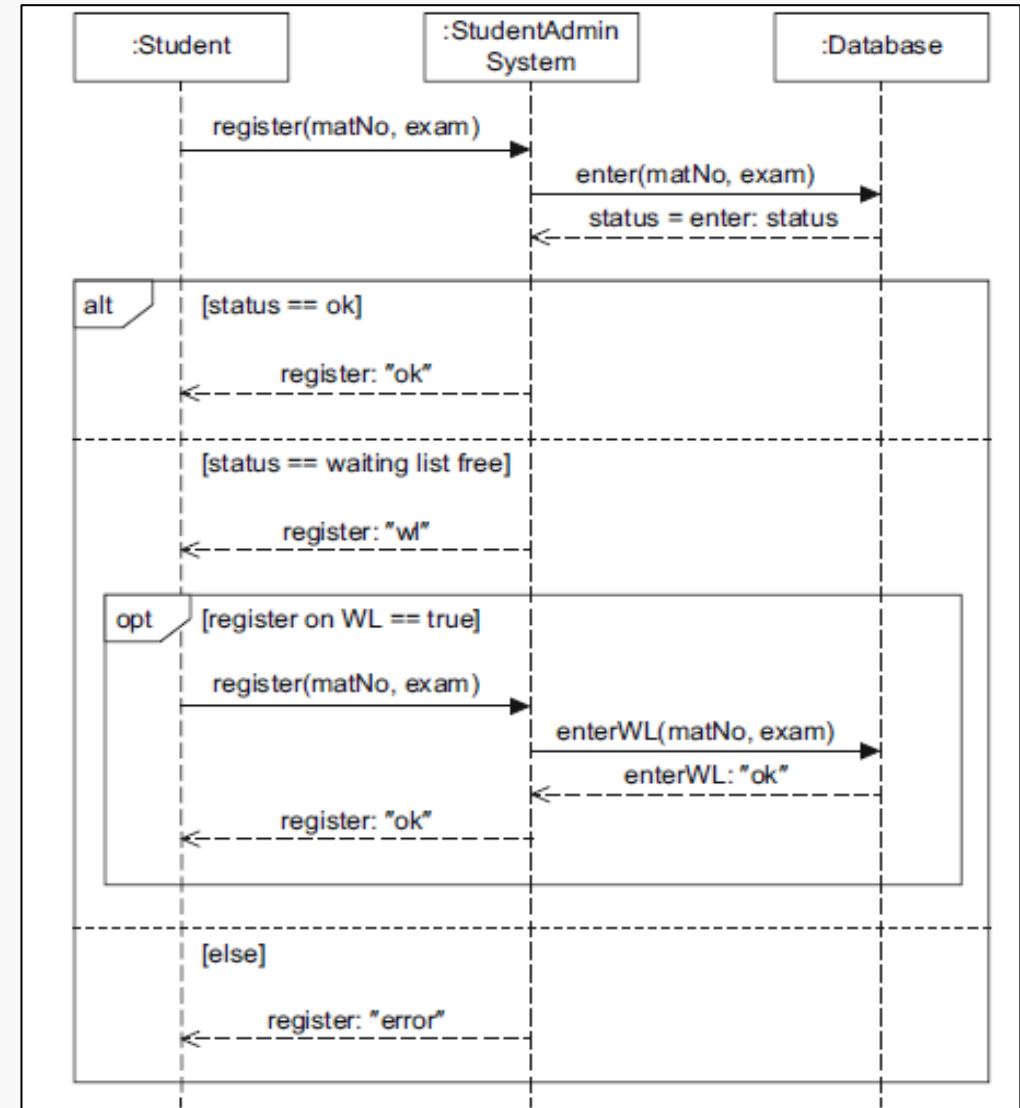
- use an **alt** fragment to represent alternative sequences
- An **alt** operator has at least two operands.
- Each operand represents an alternative path in the execution, which is equivalent for example to the *switch* statement in Java.
- Each operand has a guard.
- **Guards** are used to select the path to be executed.
- A guard is a boolean expression enclosed within square brackets.
- If there is no guard, then [true] is assumed as default value.



Drawing Sequence Diagram

Ex.: When a student wants to register for an exam, the following cases can occur:

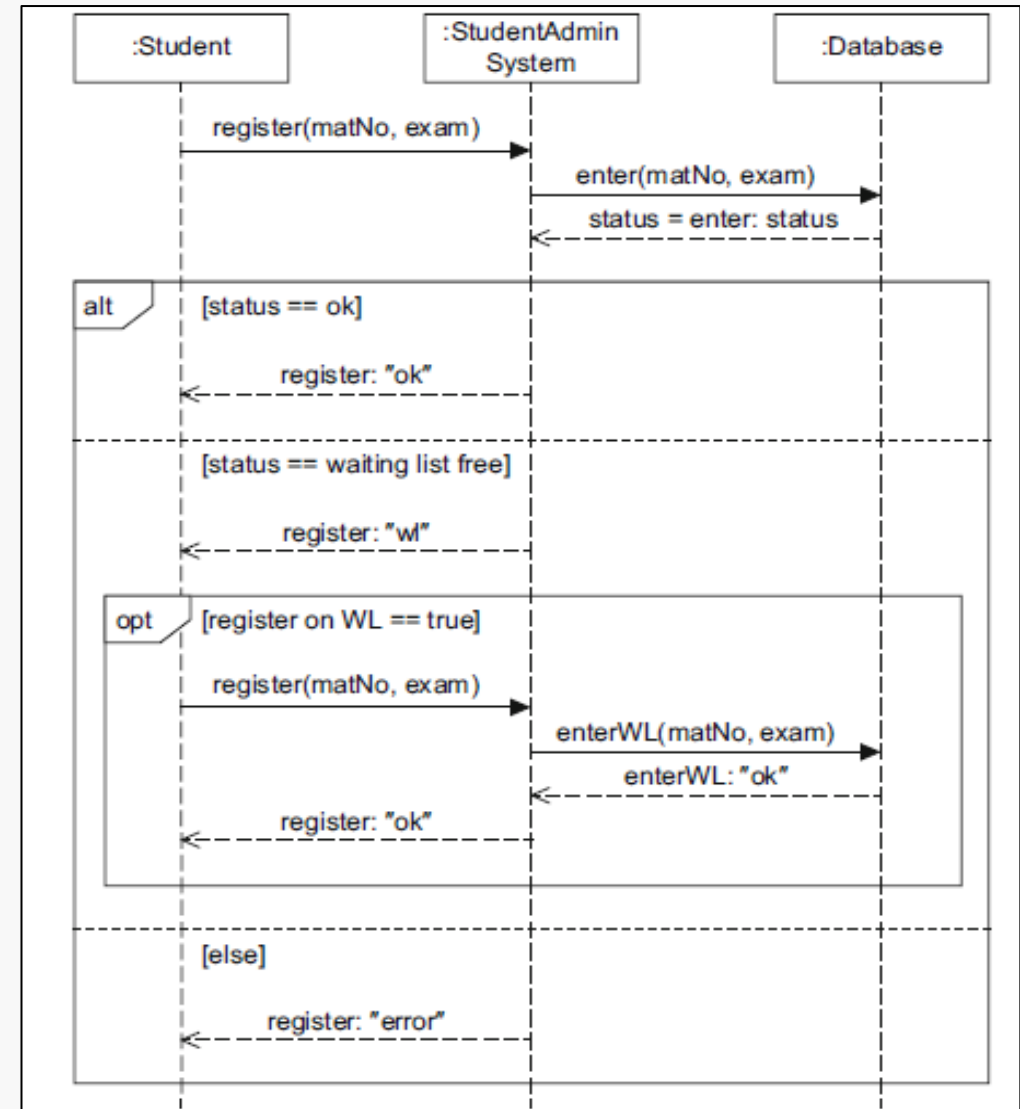
- (1) There are still places available and the student can register.*
- (2) There is a place available on the waiting list. Then the student has to decide whether to go on the waiting list.*
- (3) If there is no place available for the exam or on the waiting list for the exam, the student receives an error message and is not registered for the course.*



Drawing Sequence Diagram

The **opt** fragment corresponds to an **alt** fragment but one of which is empty.

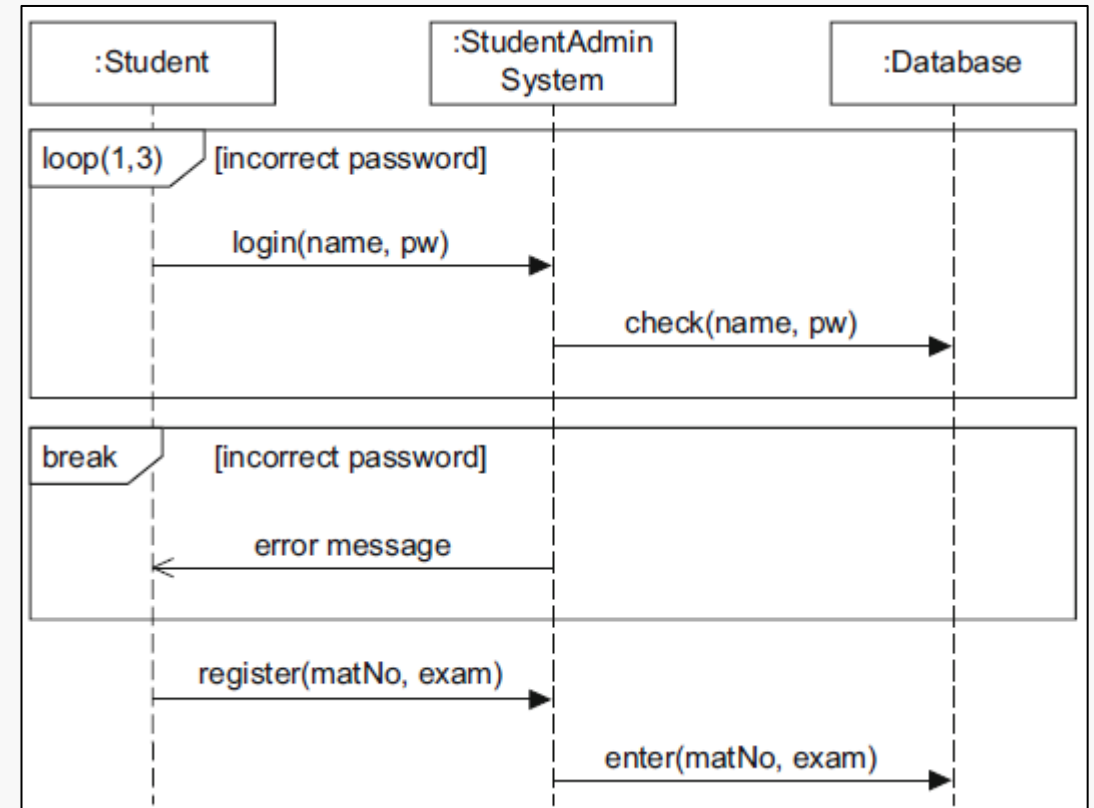
- In a programming language, this operator would be specified as an if statement without an else branch.
- *If there is a place available on the waiting list, when registering for an assignment the student can decide whether to take the place on the waiting list.*
- *If the student wants to be on the waiting list, the student has to register for it.*



Drawing Sequence Diagram

Use the **loop** fragment to express a Repeated interaction

- The keyword **loop** is given an optional specification of the number of iterations of the loop.
- It takes the form (min..max) or (min,max),
- **min** is the minimum number of iterations
- **max** is the maximum number of iterations.
- If min and max are identical, you can omit one of the two numbers and the dots.
- If there is no upper limit to the number of loop iterations, you only need to specify an asterisk *.



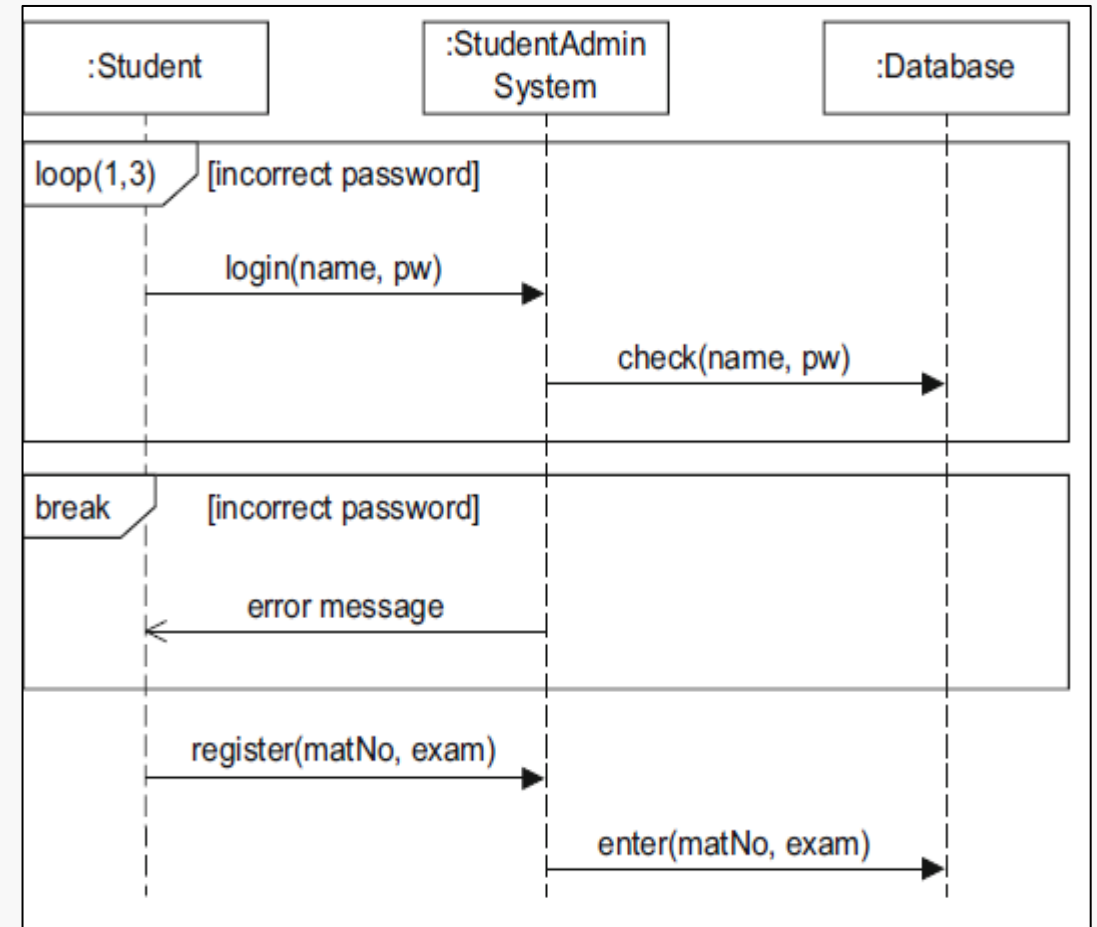
Drawing Sequence Diagram

The **break** fragment is for Exception handling

- it consists of a single operand plus a guard.

Ex.: if the password is entered incorrectly three times, the condition incorrect password is true. Thus the content of the break fragment is executed, meaning that an error message is sent to the student and the student is not allowed to register for the assignment.

So, the interaction after the end of the break fragment is skipped.



Drawing Sequence Diagram

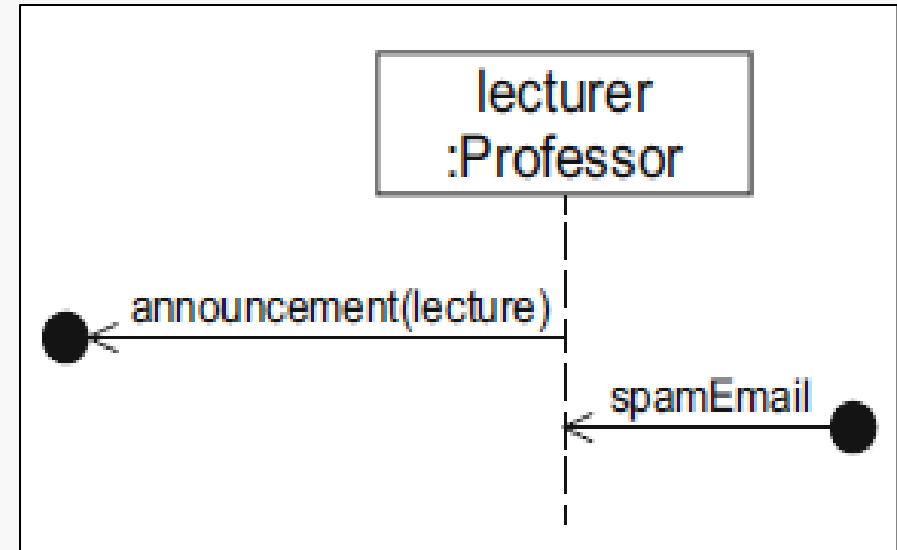
Lost and Found messages

- If the sender of a message is unknown or not relevant, use found messages.
- Use a black circle as source instead of specifying an interaction partner that sends the message.

Ex: the sender of the message spamEmail is unknown,

- If the receiver of a message is unknown or not relevant. The receiver is also noted as a black circle.

Ex.: The lecture announcement is sent to an arbitrary (unknown or irrelevant) receiver.



Example of lost and found messages